Haoyuan Zheng

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EDUCATION

University of Michigan, Ann Arbor Sep. 2026 – present

B.S.E. in Computer Engineering

Shanghai Jiao Tong University (SJTU) Sep. 2024 – Jun. 2028 (exp.)

B.S.E. in Electrical and Computer Engineering | GPA 3.68/4.0

Coursework: Data Structures and Algorithms, Honors Mathematics IV (A+), Intro to Circuits (A+), Physics II (A), Signals & Systems (A-), EECS 245, EECS 311, and EECS 370 (in progress)

Awards: SJTU Class C Excellence Scholarship (2025); Cheng Family Scholarship (2026)

RESEARCH & PROJECT EXPERIENCE

Visual SLAM and 3D Vision, IRMV, SJTU Jul. 2026 – Aug. 2026

Advisor: Prof. Hesheng Wang

- Built an object-oriented visual SLAM system in Python: SIFT/ORB features, geometric matching, relative pose estimation, PnP tracking, triangulation, and sparse bundle adjustment.
- Ran controlled experiments on frame sampling, keyframe selection, feature density, and match filtering, comparing reprojection error, point-cloud density, and runtime.
- Extended the pipeline toward dense 3DGS-SLAM with Depth Anything V2 and ADD-SLAM; diagnosed failures from limited viewpoint coverage, depth-scale inconsistency, and camera-pose drift.

Rock Pick Pose Estimation, MRover (Michigan Mars Rover Team) 2026 – present

- Developing keypoint-based pose estimation of a rock pick for autonomous retrieval: a downward-facing USB camera provides images, and the model outputs position and keypoints for the grasping subsystem.
- Built a ~10,000-frame dataset, using CoTracker for auto-labeling, and am training a YOLO-based keypoint model on it.
- Integrating with the rover software stack through ROS topics and the TF tree.

C2-MANS: Neuro-Symbolic Multi-Agent Novel Generation, SJTU Nov. 2025 – Apr. 2026

Advisor: Prof. Jiong Lou

- Designed a CrewAI pipeline of four specialized agents with Pydantic state models to keep long-form narratives logically consistent across chapters.
- Implemented structured state layers (RelationshipGraph, TimeLedger) with persistent storage for autonomous generation of 10-chapter novellas.

Space Magnetic Retrieval System Demonstrator, ENGR1000J Jul. 2025

Team project (five members)

- Designed and modeled (SolidWorks) the retrieval subsystem: spring launcher, neodymium-magnet carrier, worm gear, and motorized reel.
- Validated the design through 100 controlled trials across five distances: 1.2 m launch accuracy and a 40:1 payload-to-platform weight ratio.

SKILLS

Programming: Python, C/C++, MATLAB

Vision & Robotics: Visual SLAM, multi-view geometry, 3D reconstruction, object detection and keypoint models (YOLO), ROS, SolidWorks, prototyping

Tools: PyTorch, Pandas, Git, Linux, LaTeX